



Total Marks: 60 (20×3)

Time: **90 Minutes**

- Total no of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet
- There is no need to draw circuits in your answer sheet.

Name:	ID:	Section:
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Question 1

[20]

For this **Schmitt Trigger** circuit in Figure 1, given: $V_H = 10\text{ V}$, $V_L = -10\text{ V}$, $v_{ref} = 1\text{ V}$.

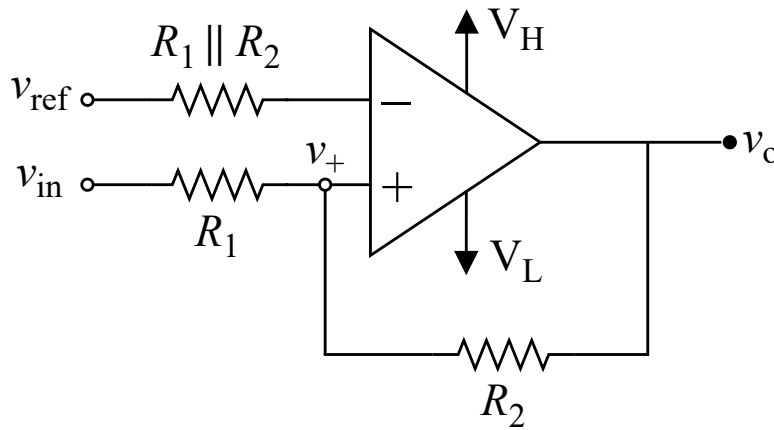


Figure 1: Schmitt Trigger

CO1	(a)	Mention the type of Schmitt trigger circuit (inverting or non-inverting)?	[2]
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For question (b), given: $R_1 = 1\text{ k}\Omega$ and $R_2 = 2\text{ k}\Omega$.

CO1	(b)	Draw the VTC (v_i vs v_o) with appropriate labels (Values of V_H , V_L , V_{TH} , V_{TL} , Hysteresis Width).	[10]
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Consider the same Figure 1. For question (c), let the input voltage be 18 V peak-to-peak sine wave.

CO1	(c)	Draw V_o vs <i>time</i> plot for the given input.	[8]
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Question 2

[20]

For the binary weighted DAC shown below, answer the following questions:

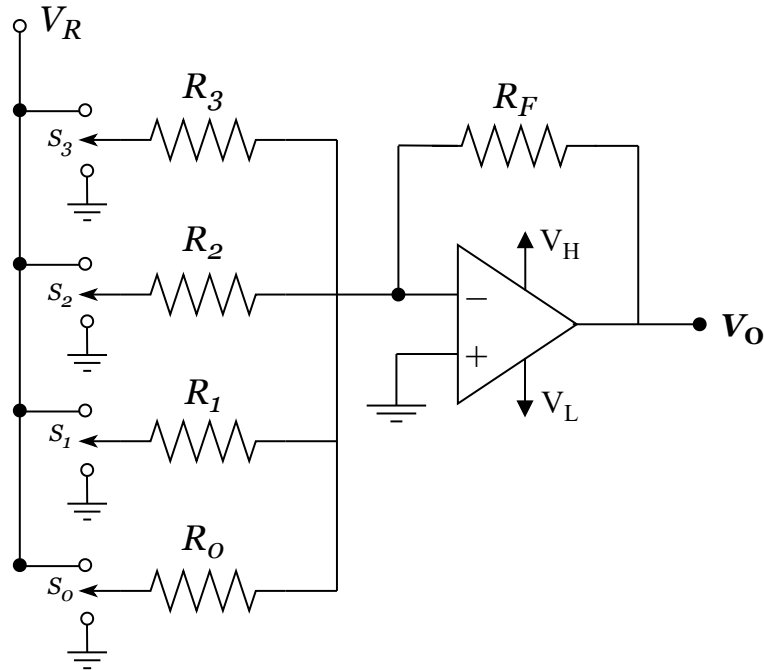


Figure 2: Binary Weighted DAC

For **part (a) & part (b)**: $V_R = -8V$, $R_3 = R_f$ Input bit sequence: $S_3S_2S_1S_0$

CO3	(a)	Show the output voltages for all possible input cases 'XXX0' in a table, where X may be 0 or 1.	[10]
CO3	(b)	Calculate the ratio of maximum output voltage and LSB(for input 0001) voltage?	[4]
CO3	(c)	Given $V_R = -8V$, find the ratio $\frac{R_3}{R_f}$, if we want the output voltage for input '1010' to be 5V?	[6]

Question 3

The NMOS NOR gate in **figure 3** is biased at $V_{DD} = 5V$. The transistor parameters are $k_n = 0.2 \text{ mA/V}^2$, $V_{TN} = 0.6V$.

For triode mode, $i_D = k_n [2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2]$, and for saturation mode, $i_D = k_n [V_{GS} - V_{TN}]^2$ [20]

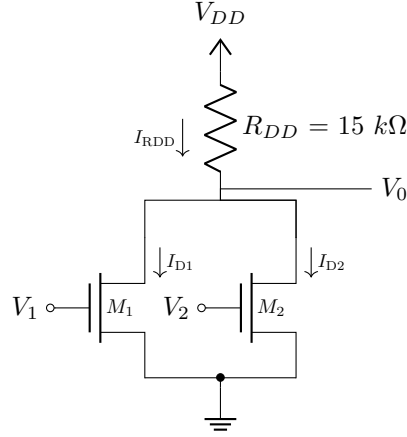


Figure 3: Circuit

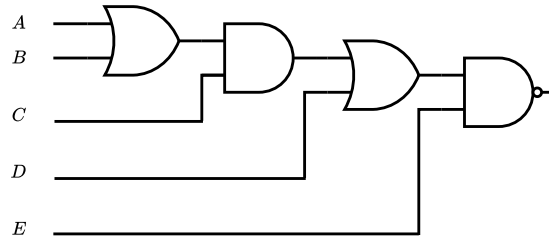


Figure 4: Boolean logic

CO1	(a)	From the above figure 3 , determine I_{RDD} , I_{D1} , I_{D2} , and V_0 for the cases where both inputs are high and both inputs are low. (Consider $5V$ as input logic high and $0V$ as input logic low.)	[8]
CO1	(b)	Determine the power dissipations for the cases where both inputs are high and both are low.	[4]
CO1	(c)	Find the boolean expression for output Y from figure 4 and design an equivalent CMOS logic circuit to implement that expression.	[8]

Question 4

For this **Triangular Wave Generator** circuit, Given: $V_H = 8\text{ V}$, $V_L = -8\text{ V}$, $R_i = 2\text{ k}\Omega$, $R_1 = 10\text{ k}\Omega$, $R_2 = 40\text{ k}\Omega$, $C = 5\text{ }\mu\text{F}$. [20]

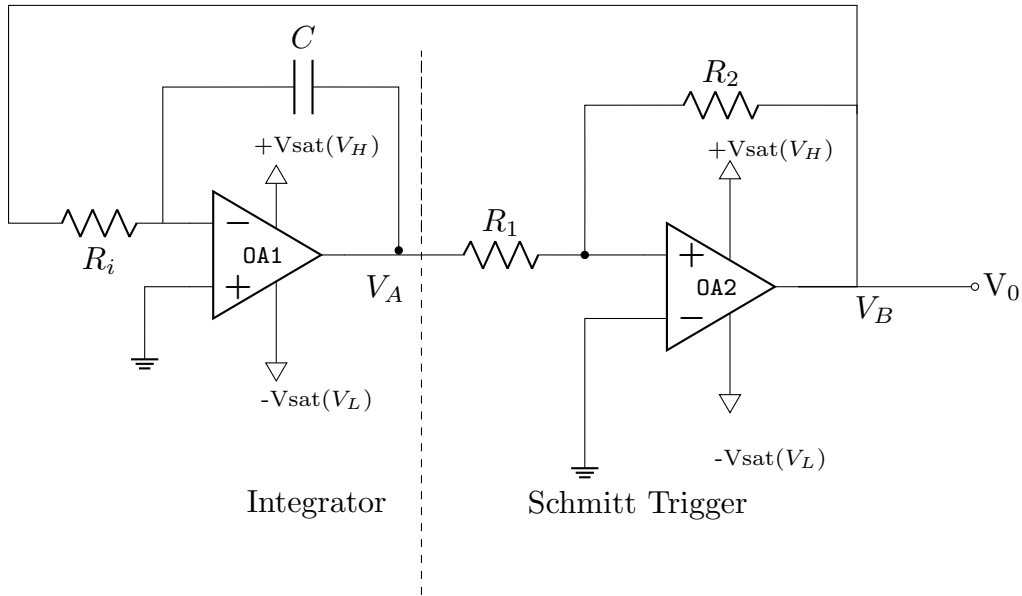


Figure 5

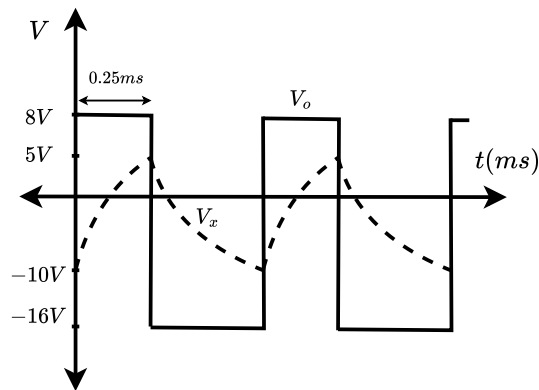


Figure 6: Output of a square wave generator circuit. Note that the dotted V_x denotes the charging and discharging of the capacitor.

CO3	(a)	Calculate the time period and the frequency of the triangular wave generated by circuit in figure 5.	[7]
CO3	(b)	Design a square wave generator that can generate a similar output waveform shown in figure 6. Draw the designed circuit with all necessary values. (Assume the value of the capacitor to be 50 nF.)	[4+4]
CO3	(c)	Find the duty cycle and frequency of the output waveform of your designed square wave generator in part (b).	[5]